Aariz Khan

AI Software Engineer

B.Tech, Artificial Intelligence & Machine Learning — CMR University, Bengaluru

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Builds AI systems end to end. A document chatbot running in production, a voice-controlled desktop agent with 103 actions, and machine learning models trained and tested on public datasets. Works in Python and TypeScript across NLP, embeddings, vector search and REST APIs.

PROJECTS

AARIZ AI — Retrieval-Augmented Document Chatbot LIVE

Next.js · TypeScript · PostgreSQL + pgvector · Supabase · Ollama · Gemini · aariz-ai-three.vercel.app · github.com/Aariz-a-z/aariz-ai

- Answers questions from your own documents and shows which sources it used.
- Reads 10 file types including scanned PDFs, splits them into chunks, and stores them as embeddings in PostgreSQL with pgvector.
- Searches by meaning and by keyword together, then reranks the results. Found the right passage 96.2% of the time across 14 test runs.

AARMA — Voice-Controlled Desktop AI Agent

Python · PySide6 (Qt 6) · SQLite · faster-whisper · edge-tts · pytest · github.com/Aariz-a-z/aarma-desktop-agent

- A Windows assistant that takes typed or spoken commands and runs 103 actions.
- Every action has its own permission setting and a risk level. Risky ones ask before they run.
- Speech recognition runs on the computer, so audio never leaves the machine. Covered by 1,932 tests.

AI Resume Analyzer LIVE

Python · Sentence-Transformers · Streamlit · PyMuPDF · aariz-ai-resume-analyzer.streamlit.app · github.com/Aariz-a-z/AI-RESUME-ANALYZER

- Compares a resume with a job description and scores how well they match, using sentence embeddings rather than keyword matching.
- Lists the skills the job asks for that the resume never mentions.

Other projects **Sales Forecasting** · **Wine Quality Classification** · **Travel Planner** · **Python Fundamentals**

TECHNICAL SKILLS

AI & Machine Learning	PyTorch, scikit-learn, regression (linear, logistic), classification (KNN, SVM), feature engineering, model evaluation (ROC-AUC, confusion matrix, precision / recall / F1, R-squared)
NLP & Retrieval	Retrieval-augmented generation, embeddings, semantic search, Sentence-Transformers, vector search (pgvector), hybrid retrieval and reranking, speech recognition (Whisper)
Data & Platforms	pandas, NumPy, exploratory data analysis, data cleaning, Matplotlib, seaborn, Plotly, Linux, Kubernetes, REST APIs, PostgreSQL, MongoDB, SQLite, Supabase
Languages & Tools	Python, TypeScript, SQL, Git, GitHub, Jupyter, Google Colab, VS Code, pytest, Streamlit, PySide6 (Qt), Next.js, React

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning · CMR University, Bengaluru — 2023 - 2027 · **CGPA 8.8**

Pre-University (MPC) · Sri Chaitanya College, Kadapa — 2021 - 2023 · **70%**

Secondary School · Keshava Reddy School, Nandyal — 2021 · **CGPA 9.8**

Languages English, Hindi, Urdu, Telugu